

## NBA 6550 Capstone Project

### Yelp Restaurant & Food Business Analysis

#### Abstract

Our project analyzes Yelp's restaurant and food business dataset to provide entry guidance for new restaurants and operational optimization strategies for existing ones. With SQLite as the core tool, we explore the correlations between geographic location, service hours, photo type, user type, check-in patterns, reviews content and restaurant performance, using star ratings and review counts as measures, based on 16,663 restaurant and food businesses data. Our outcomes indicate that moderate-competition and high-demand locations, rational service hour planning, high-quality diverse photos, elite user engagement and check-in-based operational optimization are critical to improving restaurant performance. In addition, review analysis shows that food, service, and cleanliness are the most influential factors of customer ratings, with negative experiences such as service failures, hygiene issues, and long wait times having stronger effects than positive experiences.

#### 1. Database Overview

Database (yelp\_reduced.sqlite) includes tables with structured data on Yelp businesses, users, and their interactions. The tables primarily can be linked via the keys of business\_id and user\_id. Database includes 28,450 businesses in total, 48,862 reviews (from 2006 to 2017), and 13,261 users. Additional engagement tables include 9,523 tips (from 2009 to 2017), 125,642 labeled photos, 26,881 check-in records (hour-by-day counts), and a 28,450 rows business-level count summary table. The table <sup>[1]</sup> is the basic summarization of the business table.

|   | total_restaurant_businesses | avg_stars | avg_review_count | unique_cities | unique_states | open_businesses | closed_businesses |
|---|-----------------------------|-----------|------------------|---------------|---------------|-----------------|-------------------|
| 1 | 16663                       | 3.62      | 145.09           | 388           | 24            | 13872           | 2791              |

#### 2. Business Scenario Setting

In this project, we assume that we are consultants of Yelp to analyze the dataset mainly for <sup>[2]</sup> restaurants & food businesses, provide guidance for new restaurant entrants and provide useful operational advice for existing restaurants.

We start the analysis under this scenario because it's really practical for the platform's business. When Yelp helps businesses perform better, the businesses can also reversely help Yelp attract more and more businesses to join the platform. In addition, it is also a practical application for Yelp as restaurants and food businesses make up the largest proportion of businesses <sup>[2]</sup> on Yelp, so in the subsequent operation, when they want to get more influence on the platform and let more people get into their restaurants, the businesses would pay more to the platform.

|   | Category       | count(*) |
|---|----------------|----------|
| 1 | IS FOOD        | 6872     |
| 2 | IS Restaurants | 9791     |
| 3 | Neither        | 11787    |

|   | star_rating | business_count | percentage_of_total | avg_review_count |
|---|-------------|----------------|---------------------|------------------|
| 1 | 5           | 278            | 1.67                | 58.81            |
| 2 | 4.5         | 2420           | 14.52               | 178.17           |
| 3 | 4           | 5292           | 31.76               | 192.91           |
| 4 | 3.5         | 4461           | 26.77               | 135.18           |
| 5 | 3           | 2467           | 14.81               | 97.62            |
| 6 | 2.5         | 1116           | 6.7                 | 72.44            |
| 7 | 2           | 436            | 2.62                | 46.27            |
| 8 | 1.5         | 157            | 0.94                | 25.38            |
| 9 | 1           | 36             | 0.22                | 11.78            |

Under this scenario, we first analyze the distribution of star and their average review numbers <sup>[3]</sup>. It comes out that most restaurants get 3-4.5 star, which is consistent with their average reviews. Therefore, it proves that the basis of our project works, using star and review as performance indicators. Restaurants also have a high probability of increasing their stars and reviews by improving their performance.

#### 3. Methodology

Assuming that the more reviews and higher stars, the better the restaurant. We use SQLite to finish most of the data analysis and use excel for regression analysis. The main part of our analysis is to analyze relationships between different variables and restaurants' performance.

#### 4. SQL Data Analysis

##### 4.1. Entry Strategy Analysis

##### 4.1.1. Geographic Analysis – business site selection

We assess where to open by comparing <sup>[4]</sup> restaurant density and demand signals such as review volume across locations. High review intensity indicates stronger offline traffic potential, while business concentration reflects competitive pressure. The recommended entry locations are areas with strong demand but moderate saturation, where new restaurants can gain visibility and reviews more efficiently.

The results <sup>[4]</sup> show clear differences in restaurant demand and competition across locations. Cities such as Phoenix and Toronto have a large number of highly rated restaurants, indicating strong demand but also intense competition. For example, Phoenix has 801 highly rated restaurants with an average of 235.7 reviews per business, while Toronto has 814 restaurants with 113.09 average reviews per business. In contrast, smaller cities such as Gilbert (131 restaurants, 225.62 reviews per business) and Henderson (202 restaurants, 228.33 reviews per business) show relatively high review activity compared with the number of restaurants, suggesting strong demand with lower market saturation.

|    | city       | state | high_rated_business_count | avg_star_rating | avg_review_count | review_business_count_rate |
|----|------------|-------|---------------------------|-----------------|------------------|----------------------------|
| 1  | Las Vegas  | NV    | 1430                      | 4.19            | 367.09           | 0.2567                     |
| 2  | Scottsdale | AZ    | 401                       | 4.16            | 253.16           | 0.6313                     |
| 3  | Phoenix    | AZ    | 801                       | 4.19            | 235.7            | 0.2943                     |
| 4  | Tempe      | AZ    | 195                       | 4.18            | 229.42           | 1.1765                     |
| 5  | Henderson  | NV    | 202                       | 4.19            | 228.33           | 1.1304                     |
| 6  | Gilbert    | AZ    | 131                       | 4.16            | 225.62           | 1.7223                     |
| 7  | Chandler   | AZ    | 181                       | 4.15            | 176.25           | 0.9738                     |
| 8  | Charlotte  | NC    | 418                       | 4.18            | 160.37           | 0.3837                     |
| 9  | Mesa       | AZ    | 191                       | 4.18            | 157.97           | 0.8271                     |
| 10 | Madison    | WI    | 199                       | 4.14            | 143.98           | 0.7235                     |
| 11 | Glendale   | AZ    | 105                       | 4.17            | 139.21           | 1.3258                     |
| 12 | Pittsburgh | PA    | 402                       | 4.22            | 135.35           | 0.3367                     |
| 13 | Cleveland  | OH    | 236                       | 4.22            | 127.55           | 0.5405                     |
| 14 | Toronto    | ON    | 814                       | 4.13            | 113.09           | 0.1389                     |
| 15 | Montréal   | QC    | 399                       | 4.2             | 106.41           | 0.2667                     |
| 16 | Edinburgh  | EDH   | 121                       | 4.24            | 44.74            | 0.3698                     |
| 17 | Stuttgart  | BW    | 109                       | 4.24            | 30.33            | 0.2783                     |

Considering both demand and competition, cities such as Gilbert, Glendale, Henderson, and Scottsdale appear to provide more favorable entry conditions. These locations combine strong customer engagement with a more moderate level of competition compared with highly saturated markets like Phoenix or Toronto. For new restaurants, choosing markets with high customer activity but fewer competing businesses can improve visibility and help accumulate reviews more quickly.

#### 4.1.2. Open Status Analysis – Operating hours selection

We further assess how operational decisions relate to early performance <sup>[5]</sup> outcomes. Longer and more consistent service hours increase customer access and engagement, which can accelerate review accumulation. However, extended hours also increase operational complexity and potential quality risks. Therefore, restaurants should align service hours with local demand patterns, like lunch and dinner peaks to ensure best benefits themselves.

The result <sup>[5]</sup> shows moderate seasonality in monthly review activity. Review counts peak in summer months such as July and August and decline in winter months like December, while average ratings remain relatively stable

|    | review_month | total_reviews | avg_review_rating |
|----|--------------|---------------|-------------------|
| 1  | 08           | 3273          | 3.7               |
| 2  | 07           | 3232          | 3.72              |
| 3  | 09           | 2959          | 3.68              |
| 4  | 06           | 2907          | 3.73              |
| 5  | 05           | 2699          | 3.71              |
| 6  | 10           | 2624          | 3.72              |
| 7  | 03           | 2601          | 3.73              |
| 8  | 04           | 2587          | 3.73              |
| 9  | 11           | 2568          | 3.71              |
| 10 | 01           | 2522          | 3.72              |
| 11 | 02           | 2507          | 3.7               |
| 12 | 12           | 2160          | 3.74              |

throughout the year. Also, from result <sup>[6]</sup>, <sup>[6]</sup> restaurants operating 12–18 hours per day receive the highest average review counts, indicating that longer service availability increases customer exposure and traffic. In contrast, businesses with shorter operating hours (1–5 hours) tend to receive

|   | avg_daily_hours_tier        | business_count | avg_star_rating | avg_review_count |
|---|-----------------------------|----------------|-----------------|------------------|
| 1 | 0 Hours (Null/No Hour Data) | 2550           | 3.22            | 73.1             |
| 2 | 1-5 Hours                   | 1447           | 3.92            | 166.22           |
| 3 | 6-12 Hours                  | 9105           | 3.76            | 152.88           |
| 4 | 12-18 Hours                 | 3425           | 3.46            | 169.53           |
| 5 | ≥19 Hours                   | 136            | 3.24            | 133.43           |

higher ratings, while restaurants operating nearly 24 hours (≥19 hours) show lower ratings and only moderate review volumes, suggesting that extremely long service hours do not necessarily improve customer satisfaction.

Overall, service availability and seasonal demand both influence restaurant performance. Moderate-to-long operating hours help increase customer traffic, while extremely long service hours may dilute service quality. Since customer demand is higher during summer months, restaurants may benefit from adjusting or extending operating hours during peak seasons to capture additional customer traffic and improve visibility.

## 4.2. Business Operation Improvement Analysis

### 4.2.1. Photos Analysis – marketing of photos quantity and combinations

We evaluate how photo volume and photo types relate to performance outcomes. Photos serve as a visibility and conversion tool as richer visual content can reduce uncertainty and attract first-time visits. Operationally, restaurants should maintain up-to-date, high-quality photos to improve click-to-visit conversion and support review acquisition.

The results show a clear relationship between the number [7] and types of photos and restaurant performance. Businesses with more photos tend to receive significantly more customer engagement [7]. Restaurants with only 1–9 photos receive an average of 117.5 reviews and an average rating of 3.62, while those with more than 100 photos receive an average of 1,405 reviews and an average rating of 4.03. This indicates that higher photo availability is associated with both higher review counts and slightly higher ratings. In addition, food photos represent the largest share of all images (79,032), followed by inside and outside photos, suggesting that customers are most interested in visual information about dishes [8]. When different types of photos are combined, restaurants often achieve higher ratings, particularly when food photos are paired with other categories such as drinks, menus, or interior images [9].

|   | user_photo_count_tier | business_count | avg_star_rating | avg_review_count |
|---|-----------------------|----------------|-----------------|------------------|
| 1 | 1-9 Photos            | 7690           | 3.62            | 117.5            |
| 2 | 10-49 Photos          | 2952           | 3.78            | 300.07           |
| 3 | 50-99 Photos          | 261            | 3.98            | 722.43           |
| 4 | ≥100 Photos           | 101            | 4.03            | 1405.42          |

|   | photo_label | photo_count | avg_business_stars |
|---|-------------|-------------|--------------------|
| 1 | inside      | 28517       | 3.83               |
| 2 | food        | 79032       | 3.83               |
| 3 | drink       | 4155        | 3.83               |
| 4 | menu        | 410         | 3.82               |
| 5 | outside     | 10173       | 3.78               |

|    | combo_size | label1  | label2  | label3  | business_count | avg_stars |
|----|------------|---------|---------|---------|----------------|-----------|
| 1  | 1          | food    |         |         | 8750           | 3.83      |
| 2  | 1          | inside  |         |         | 7156           | 3.83      |
| 3  | 1          | outside |         |         | 4569           | 3.78      |
| 4  | 1          | drink   |         |         | 2096           | 3.83      |
| 5  | 1          | menu    |         |         | 312            | 3.82      |
| 6  | 2          | food    | inside  |         | 5513           | 3.95      |
| 7  | 2          | food    | outside |         | 3485           | 3.97      |
| 8  | 2          | inside  | outside |         | 3282           | 3.85      |
| 9  | 2          | drink   | food    |         | 1935           | 3.98      |
| 10 | 2          | drink   | inside  |         | 1719           | 3.92      |
| 11 | 2          | drink   | outside |         | 1170           | 3.93      |
| 12 | 2          | food    | menu    |         | 275            | 3.99      |
| 13 | 2          | inside  | menu    |         | 231            | 3.95      |
| 14 | 2          | menu    | outside |         | 156            | 3.91      |
| 15 | 2          | drink   | menu    |         | 98             | 3.9       |
| 16 | 3          | food    | inside  | outside | 2756           | 3.92      |
| 17 | 3          | drink   | food    | inside  | 1610           | 3.93      |
| 18 | 3          | drink   | food    | outside | 1100           | 4.05      |
| 19 | 3          | drink   | inside  | outside | 1060           | 3.92      |
| 20 | 3          | food    | inside  | menu    | 215            | 3.87      |
| 21 | 3          | food    | menu    | outside | 150            | 4.04      |
| 22 | 3          | inside  | menu    | outside | 136            | 3.9       |
| 23 | 3          | drink   | food    | menu    | 97             | 4.04      |
| 24 | 3          | drink   | inside  | menu    | 96             | 3.86      |
| 25 | 3          | drink   | menu    | outside | 71             | 4.03      |

Therefore, photos are an important tool for attracting customer attention and increasing engagement. For newly opened restaurants, it is recommended to upload a large number of high-quality photos early to improve visibility and encourage customer interaction. Restaurants should prioritize food photos while also including interior, drink, and menu images to provide more complete information. Maintaining a diverse photo gallery can help build customer trust, attract more reviews, and support stronger rating performance.

#### 4.2.2. Elite Analysis – capitalizing on the influence of elite users

We analyze whether Yelp Elite users are associated with higher-quality engagement, such as more informative reviews (e.g., higher “useful” votes) and stronger signaling effects on reputation. Elite reviews can amplify credibility and discovery, making them valuable for reputation building.

Table [10] indicate that Elite users have a “reputation repair” effect on low-rated restaurants as in 2.5-star and 3-star establishments, their average ratings are significantly higher than those of regular users. For example, the gap between regular user and elite user in 2.5-star restaurants is 0.37, proving that they can effectively boost merchant reputation. Conversely, high-rated restaurants (4.5 stars and above) rely on regular users for “reputation amplification,” as these users dominate review volume.

|    | restaurant_star_rating | user_type    | review_count | avg_review_rating |
|----|------------------------|--------------|--------------|-------------------|
| 1  | 5                      | Regular User | 320          | 4.89              |
| 2  | 5                      | Elite User   | 68           | 4.78              |
| 3  | 4.5                    | Regular User | 4381         | 4.42              |
| 4  | 4.5                    | Elite User   | 1064         | 4.37              |
| 5  | 4                      | Regular User | 9669         | 3.99              |
| 6  | 4                      | Elite User   | 2654         | 4.02              |
| 7  | 3.5                    | Regular User | 6365         | 3.5               |
| 8  | 3.5                    | Elite User   | 1905         | 3.58              |
| 9  | 3                      | Regular User | 2948         | 2.94              |
| 10 | 3                      | Elite User   | 942          | 3.26              |
| 11 | 2.5                    | Regular User | 1233         | 2.42              |
| 12 | 2.5                    | Elite User   | 341          | 2.79              |
| 13 | 2                      | Regular User | 443          | 1.81              |
| 14 | 2                      | Elite User   | 96           | 2.64              |
| 15 | 1.5                    | Regular User | 152          | 1.48              |
| 16 | 1.5                    | Elite User   | 22           | 1.73              |
| 17 | 1                      | Regular User | 30           | 1.1               |
| 18 | 1                      | Elite User   | 6            | 1.33              |

While Elite users are scarce, accounting for only 6.05% of all reviewers, their influence is still substantial: they have a higher average total review count (80.54) and average number of fans (6.39), making each of their reviews more impactful and far-reaching [11]. Restaurants can benefit by ensuring consistent service quality and encouraging shareable experiences that increase the likelihood of Elite engagement. Low-rated establishments should proactively attract Elite users to leverage their influence for reputation repair, while high-rated ones should maintain service standards, nurture Elite relationships for stronger brand endorsement, and treat Elite users as critical assets to convert their influence into long-term reputation-building assets.

|   | total_restaurant_reviewers | elite_user_count | elite_user_percentage | avg_user_total_reviews | avg_user_rating | avg_user_fans |
|---|----------------------------|------------------|-----------------------|------------------------|-----------------|---------------|
| 1 | 9504                       | 575              | 6.05                  | 80.54                  | 3.73            | 6.39          |

#### 4.2.3. Check-in Analysis – peak-hour and operating strategy

We use check-in patterns by week of day and hour to infer peak demand windows and operational bottlenecks. High check-in intensity during specific time blocks can indicate staffing and capacity stress points that affect customer experience and ratings.



Our analysis shows that check-ins are strongly correlated with business performance. Restaurants with check-in records have a far higher average number of reviews (146.38) than those without (10.97), making check-ins a key indicator of user engagement and store visibility<sup>[12]</sup>. Weekends emerge as peak traffic periods, with Sunday (1,074,787 check-ins) and Saturday (927,762 check-ins) leading in volume, while evening hours (18:00–21:00) represent the absolute peak, with 512,798 check-ins at 18:00. Notably, high check-in volumes at 2:00 and 23:00 also reflect strong demand in the late-night dining market. Weekday patterns are similarly clear that check-in peaks still occur in the evening (18:00–21:00), though overall volume is lower than on weekends<sup>[13],[14],[15]</sup>.

Restaurants can use these insights to optimize labor scheduling, reduce waiting times, and align promotions with off-peak periods to smooth demand and improve overall performance. This includes increasing service staff during weekend and evening peaks to avoid experience declines, launching off-peak promotions (e.g., weekday lunchtime) to balance traffic, and refining service processes during high-demand periods to ensure consistent, high-quality experiences that convert peak traffic into positive reviews.

### 4.3. Review Analysis – the 3 key dimensions to avoid failures and drive success

| review_count_tier | business_count | avg_star_rating | avg_review_count |
|-------------------|----------------|-----------------|------------------|
| 1 1-9             | 1209           | 3.42            | 5.8              |
| 2 10-49           | 5044           | 3.48            | 27.83            |
| 3 50-100          | 3684           | 3.6             | 72.56            |
| 4 100-199         | 3305           | 3.71            | 142.87           |
| 5 200-299         | 1463           | 3.8             | 243.23           |
| 6 300-399         | 782            | 3.87            | 344.84           |
| 7 400-499         | 380            | 3.85            | 444.45           |
| 8 500+            | 796            | 3.94            | 925.09           |

Finally, we do two analyses on user reviews, for better support the restaurant-oriented business decisions. The first table summarizes the effectiveness of analyzing the reviews, that businesses with more reviews generally receive higher average star ratings<sup>[16]</sup>. It reflects the number of reviews itself may reduce information asymmetry, allowing subsequent customers to form more stable expectations about ratings.

Then, we apply a multiple linear regression approach to examine how different review tags affect star ratings, with the goal of identifying the key drivers of business performance improvement. We classify review text into several dimensions, including service, price, wait time, cleanliness, food, value, and environment, and further divide each category into positive and negative expressions (detailed words see appendix). We firstly get the panel<sup>[17]</sup>, and we validate the accuracy of the tags<sup>[18]</sup> and find that reviews with positive tags consistently have higher average star ratings than those with corresponding negative tags. The largest gaps appear in service (2.494) and cleanliness (1.856), followed by food (1.389). The result confirms that the signed tags have strong discriminating power in separating high-rated and low-rated reviews.

We then run the regression in Excel and obtain the results shown in

|   | checkin_status | business_count | avg_stars | avg_reviews |
|---|----------------|----------------|-----------|-------------|
| 1 | 0 Check-Ins    | 158            | 3.66      | 10.97       |
| 2 | Has Check-Ins  | 16505          | 3.62      | 146.38      |

|   | weekday   | weekday_sum |
|---|-----------|-------------|
| 1 | Sunday    | 1074787     |
| 2 | Saturday  | 927762      |
| 3 | Friday    | 685552      |
| 4 | Thursday  | 630002      |
| 5 | Wednesday | 571943      |
| 6 | Monday    | 382143      |
| 7 | Tuesday   | 357853      |

|    | hour  | hour_sum |
|----|-------|----------|
| 1  | 18:00 | 512798   |
| 2  | 19:00 | 462656   |
| 3  | 2:00  | 452596   |
| 4  | 23:00 | 446805   |
| 5  | 21:00 | 421559   |
| 6  | 0:00  | 390609   |
| 7  | 20:00 | 360888   |
| 8  | 1:00  | 323055   |
| 9  | 17:00 | 310839   |
| 10 | 16:00 | 188340   |
| 11 | 3:00  | 188130   |
| 12 | 15:00 | 128339   |
| 13 | 22:00 | 122955   |
| 14 | 4:00  | 121775   |
| 15 | 14:00 | 84200    |
| 16 | 6:00  | 47624    |
| 17 | 13:00 | 36944    |
| 18 | 12:00 | 14616    |
| 19 | 10:00 | 9233     |
| 20 | 11:00 | 6081     |

|    | weekday   | hour  | sum_val |
|----|-----------|-------|---------|
| 1  | Monday    | 19:00 | 80712   |
| 2  | Monday    | 20:00 | 65356   |
| 3  | Monday    | 18:00 | 58853   |
| 4  | Monday    | 23:00 | 56761   |
| 5  | Monday    | 4:00  | 42009   |
| 6  | Monday    | 17:00 | 37116   |
| 7  | Monday    | 15:00 | 20503   |
| 8  | Monday    | 14:00 | 13404   |
| 9  | Monday    | 12:00 | 4371    |
| 10 | Monday    | 11:00 | 3058    |
| 11 | Tuesday   | 23:00 | 59024   |
| 12 | Tuesday   | 20:00 | 58790   |
| 13 | Tuesday   | 18:00 | 56530   |
| 14 | Tuesday   | 21:00 | 48256   |
| 15 | Tuesday   | 4:00  | 37714   |
| 16 | Tuesday   | 17:00 | 33670   |
| 17 | Tuesday   | 16:00 | 29715   |
| 18 | Tuesday   | 15:00 | 21040   |
| 19 | Tuesday   | 13:00 | 8573    |
| 20 | Tuesday   | 12:00 | 4541    |
| 21 | Wednesday | 1:00  | 90266   |
| 22 | Wednesday | 2:00  | 86303   |
| 23 | Wednesday | 19:00 | 78605   |
| 24 | Wednesday | 0:00  | 75581   |

|   | dimension   | pos_count | pos_avg | neg_count | neg_avg | avg_star_gap |
|---|-------------|-----------|---------|-----------|---------|--------------|
| 1 | service     | 2632      | 4.248   | 587       | 1.755   | 2.494        |
| 2 | cleanliness | 1474      | 3.981   | 918       | 2.125   | 1.856        |
| 3 | food        | 11444     | 4.18    | 3464      | 2.792   | 1.389        |
| 4 | wait        | 496       | 3.966   | 2809      | 2.763   | 1.203        |
| 5 | value       | 969       | 4.027   | 195       | 2.944   | 1.083        |
| 6 | price       | 2506      | 3.972   | 2249      | 3.006   | 0.966        |
| 7 | environment | 1802      | 4.126   | 1966      | 3.578   | 0.548        |

| review_id                   | business_id | stars      | date                   | text | Year | service | service | price | price | wait | wait | clean | clean | food | food | value | value | enviro | enviro |
|-----------------------------|-------------|------------|------------------------|------|------|---------|---------|-------|-------|------|------|-------|-------|------|------|-------|-------|--------|--------|
| F05xwV6oa7uPFCpBrIGAnz      | 4           | 2017/3/11  | Myself, sister and h   | 2017 | 0    | 0       | 0       | 0     | 0     | 0    | 0    | 0     | 0     | 0    | 1    | 1     | 0     | 0      | 0      |
| uFUFvmapYm5fEaMcGZaQuG1c    | 5           | 2014/8/10  | Looking for ice crea   | 2014 | 1    | 0       | 0       | 0     | 0     | 0    | 0    | 0     | 0     | 0    | 0    | 0     | 0     | 0      | 0      |
| zVD0p9nRgG-iGu-e3NQCj3Mj    | 5           | 2017/2/18  | I love their Mango     | 2017 | 0    | 0       | 0       | 0     | 0     | 0    | 0    | 0     | 0     | 0    | 1    | 0     | 0     | 0      | 0      |
| hw8qfRfRAUfXhK5FvER1eVnq    | 3           | 2017/2/1   | Yes, bohemian? Is i    | 2017 | 0    | 0       | 0       | 0     | 0     | 1    | 0    | 0     | 0     | 0    | 0    | 0     | 0     | 0      | 0      |
| SHvooaOMH541RVaZc89cnuPR    | 4           | 2017/1/13  | Nice place to hang     | 2017 | 0    | 0       | 0       | 0     | 0     | 0    | 0    | 0     | 0     | 0    | 0    | 0     | 0     | 0      | 0      |
| FHSUc0DBR1yp1WBAuQg81ke     | 4           | 2013/2/23  | Best skinny little fri | 2013 | 0    | 0       | 0       | 0     | 0     | 0    | 0    | 0     | 0     | 0    | 0    | 0     | 0     | 0      | 0      |
| yf4G76dU9Roc1WBAuQg81ke     | 5           | 2016/12/10 | This place rocks!!!    | 2016 | 0    | 0       | 0       | 0     | 0     | 0    | 0    | 0     | 0     | 0    | 1    | 0     | 0     | 0      | 0      |
| 139w-oppdy6vN7V4cD-KqgE3O   | 3           | 2014/8/10  | First time in a long   | 2014 | 0    | 0       | 0       | 0     | 0     | 0    | 0    | 0     | 0     | 0    | 0    | 0     | 0     | 0      | 0      |
| M9ADNvXZm2n7V4cD-KqgE3O     | 3           | 2017/9/6   | I like going here, b   | 2017 | 0    | 0       | 0       | 0     | 0     | 0    | 0    | 0     | 0     | 0    | 0    | 0     | 0     | 0      | 0      |
| hK1NLeqPZf1n7V4cD-KqgE3O    | 1           | 2016/5/22  | Brought my family      | 2016 | 0    | 0       | 0       | 0     | 0     | 0    | 0    | 0     | 0     | 0    | 0    | 0     | 0     | 0      | 0      |
| h1HTKzNmLz:n7V4cD-KqgE3O    | 3           | 2016/7/13  | 6/29We came to G       | 2016 | 0    | 0       | 0       | 0     | 0     | 0    | 0    | 0     | 0     | 0    | 0    | 0     | 0     | 0      | 0      |
| Gv3rr6Y5B0Kc:PA_Lohse-hob   | 4           | 2016/1/17  | My favorite fast fox   | 2016 | 0    | 0       | 0       | 0     | 0     | 0    | 0    | 0     | 0     | 0    | 0    | 0     | 0     | 0      | 0      |
| 7hRc_4lUiaZ5Y BSEZEDH6AVDk  | 4           | 2016/1/3   | Great burgers, nice    | 2016 | 1    | 0       | 0       | 0     | 0     | 0    | 0    | 0     | 0     | 0    | 0    | 0     | 0     | 0      | 0      |
| db98frNokocVdPZgZNAu8t8i    | 2           | 2015/11/12 | Good food but way      | 2015 | 0    | 0       | 1       | 0     | 0     | 0    | 0    | 0     | 0     | 0    | 1    | 0     | 0     | 0      | 0      |
| hgm1Bz0dG1cVdPZgZNAu8t8i    | 3           | 2016/6/23  | To start: Incredible   | 2016 | 0    | 0       | 0       | 0     | 0     | 0    | 0    | 0     | 0     | 0    | 0    | 0     | 0     | 0      | 1      |
| gPp2ym_HBUU1nH9BPX0FB8k     | 1           | 2011/10/8  | This place is just a   | 2011 | 0    | 0       | 0       | 0     | 0     | 0    | 0    | 0     | 0     | 0    | 0    | 0     | 0     | 0      | 0      |
| pl5NvjrhDHPFvexk8rGY_RIPp   | 5           | 2017/6/25  | My husband and i s     | 2017 | 0    | 0       | 0       | 0     | 0     | 0    | 0    | 0     | 0     | 0    | 1    | 0     | 0     | 0      | 0      |
| GA0303ohd9K5:5q6Xh-Ucia78b  | 4           | 2015/12/26 | I'm from the Outer     | 2015 | 0    | 0       | 0       | 0     | 0     | 0    | 0    | 0     | 0     | 0    | 1    | 0     | 1     | 0      | 0      |
| mYul1cVmAmdz:5q6Xh-Ucia78b  | 2           | 2016/8/20  | Was in Charlotte fo    | 2016 | 0    | 0       | 0       | 0     | 0     | 0    | 0    | 0     | 0     | 0    | 0    | 0     | 0     | 0      | 0      |
| QvTZFa8hB_gE5pOyH4Fon5783   | 2           | 2012/7/30  | Went to Apple Wei      | 2012 | 0    | 0       | 0       | 0     | 0     | 0    | 0    | 0     | 0     | 0    | 0    | 0     | 0     | 0      | 0      |
| m7u7CD-VhwW5d7AQMDYdJ4K     | 5           | 2016/9/17  | Beau petit bijou qu    | 2016 | 0    | 0       | 0       | 0     | 0     | 0    | 0    | 0     | 0     | 0    | 0    | 0     | 0     | 0      | 0      |
| 4hZDqHv8XvnmLP:jlldoaYm7    | 5           | 2014/4/11  | This is by far the be  | 2014 | 0    | 0       | 0       | 0     | 0     | 0    | 0    | 0     | 0     | 0    | 0    | 0     | 0     | 0      | 0      |
| aZHuDUUNSR5f:mLP:jlldoaYm7  | 5           | 2010/12/12 | I love Napoli's. Oh l  | 2010 | 0    | 0       | 0       | 0     | 0     | 0    | 0    | 0     | 0     | 0    | 0    | 0     | 0     | 0      | 0      |
| CqG5-nloXAN_lykxWKGQW5      | 4           | 2015/7/1   | ok, so i went to Zor   | 2015 | 0    | 0       | 0       | 0     | 0     | 0    | 0    | 0     | 0     | 0    | 0    | 0     | 0     | 0      | 0      |
| rw2HfRhoTkvM3oV993ED5pYr    | 5           | 2017/11/1  | Wow. So good. The      | 2017 | 0    | 0       | 1       | 0     | 0     | 0    | 1    | 0     | 0     | 0    | 1    | 0     | 0     | 0      | 0      |
| cx7Y3n-aP5SLNvHwZpgGTURc    | 3           | 2016/5/14  | The service here is    | 2016 | 0    | 0       | 0       | 0     | 0     | 0    | 0    | 0     | 0     | 0    | 0    | 1     | 0     | 0      | 0      |
| mu2QutHqPvRhZmFvKcamE       | 4           | 2017/8/5   | We didn't have to l    | 2017 | 0    | 0       | 0       | 0     | 0     | 0    | 0    | 0     | 0     | 0    | 0    | 0     | 0     | 0      | 0      |
| cnVnShd4Hh:O_UJC_1dXkAmki   | 3           | 2016/9/5   | Went today for an      | 2016 | 0    | 0       | 0       | 0     | 0     | 0    | 0    | 0     | 0     | 0    | 1    | 0     | 0     | 0      | 0      |
| ICq4d14YdGn:W2hNHL9F0Ww     | 2           | 2017/7/8   | I love the milkshak    | 2017 | 0    | 1       | 0       | 0     | 0     | 0    | 0    | 0     | 0     | 0    | 1    | 0     | 0     | 0      | 0      |
| q66YACgY3c3C:W2hNHL9F0Ww    | 1           | 2016/3/13  | Worst drive-thru se    | 2016 | 0    | 0       | 0       | 0     | 0     | 0    | 0    | 0     | 0     | 0    | 0    | 0     | 0     | 0      | 0      |
| qyzQ2zG6BfRf:ww_MXk-qdLeYr  | 5           | 2012/12/31 | Pho Linh in hands c    | 2012 | 0    | 0       | 0       | 0     | 0     | 0    | 0    | 0     | 0     | 0    | 1    | 0     | 0     | 0      | 0      |
| vGFC_Qbqepv:ww_MXk-qdLeYr   | 4           | 2017/1/4   | Great Vietnamese f     | 2016 | 0    | 0       | 0       | 0     | 0     | 0    | 0    | 0     | 0     | 0    | 0    | 0     | 0     | 0      | 0      |
| P15PCP-xx7DhD:U1ZjgF-kfuv_r | 5           | 2016/11/3  | So happy with this     | 2016 | 0    | 0       | 0       | 0     | 0     | 0    | 0    | 0     | 0     | 0    | 1    | 0     | 0     | 0      | 0      |
| tpMc_MyIa98X:U1ZjgF-kfuv_r  | 5           | 2016/11/22 | The food was GREA      | 2016 | 0    | 0       | 0       | 0     | 0     | 0    | 0    | 0     | 0     | 0    | 0    | 0     | 0     | 0      | 0      |
| LgLeV7Z2G65:3b4nqfG6GOrLQ   | 5           | 2012/12/28 | I am writing this r    | 2012 | 0    | 1       | 0       | 0     | 0     | 0    | 0    | 0     | 0     | 0    | 0    | 0     | 0     | 0      | 0      |
| UtxyePntYfYmY:dPZ9vKt5dVf   | 5           | 2017/7/17  | Wow wow wow. Ai        | 2017 | 1    | 0       | 0       | 0     | 0     | 0    | 0    | 0     | 0     | 0    | 1    | 0     | 0     | 0      | 0      |
| YI_XGIXwXVOL:dPZ9vKt5dVf    | 5           | 2017/7/26  | We loved Montress      | 2017 | 0    | 0       | 0       | 0     | 0     | 0    | 0    | 0     | 0     | 0    | 0    | 0     | 0     | 0      | 0      |
| XTX0H4Y_C02hZ:dPZ9vKt5dVf   | 5           | 2017/2/5   | fantastic food in ou   | 2017 | 1    | 0       | 0       | 0     | 0     | 0    | 0    | 0     | 0     | 0    | 0    | 0     | 0     | 0      | 0      |
| yc15MULb04Sjg:dPZ9vKt5dVf   | 5           | 2017/1/14  | Truly a wonderful c    | 2017 | 1    | 0       | 0       | 0     | 0     | 0    | 0    | 0     | 0     | 0    | 0    | 0     | 0     | 0      | 0      |



Table <sup>[19]</sup>. Since we believe the review is affected by multiple factors, all variables are included in the same model. After controlling for year, the overall model remains highly statistically significant (Significance  $F \approx 0$ ,  $R^2 = 0.231$ ,  $p$ -value  $< 0.05$ ). The results strongly suggest that service, cleanliness, and food are the most important factors to forecast high-rated and low-rated reviews, and that the absolute magnitudes of the negative coefficients are generally larger than those of the positive coefficients. In particular, food\_pos has the strongest positive effect (+0.646), while service\_neg (-1.531) and cleanliness\_neg (-1.278) both have the strongest negative effects.

Overall, restaurant ratings are not only positively driven by food quality, but are even more strongly affected by negative experiences such as service failures, hygiene issues, and long waiting times. This also implies that, in practice, avoiding serious negative experiences is often more important than creating additional positive surprises in protecting overall rating performance.

[19]

| SUMMARY OUTPUT        |              |                |              |             |                |              |
|-----------------------|--------------|----------------|--------------|-------------|----------------|--------------|
| Regression Statistics |              |                |              |             |                |              |
| Multiple R            | 0.481020645  |                |              |             |                |              |
| R Square              | 0.231380861  |                |              |             |                |              |
| Adjusted R Square     | 0.231027451  |                |              |             |                |              |
| Standard Error        | 1.208208055  |                |              |             |                |              |
| Observations          | 32639        |                |              |             |                |              |
| ANOVA                 |              |                |              |             |                |              |
|                       | df           | SS             | MS           | F           | Significance F |              |
| Regression            | 15           | 14335.85461    | 955.7236408  | 654.709851  | 0              |              |
| Residual              | 32623        | 47621.96916    | 1.459766703  |             |                |              |
| Total                 | 32638        | 61957.82377    |              |             |                |              |
|                       | Coefficients | Standard Error | t Stat       | P-value     | Lower 95.0%    | Upper 95.0%  |
| Intercept             | -32.83273882 | 6.109967284    | -5.373635781 | 7.76929E-08 | -44.80849896   | -20.85697868 |
| Year Control          | 0.018128827  | 0.003032752    | 5.977682939  | 2.28681E-09 | 0.012184523    | 0.024073132  |
| service_pos           | 0.369234978  | 0.02490223     | 14.82738634  | 1.41384E-49 | 0.320425694    | 0.418044261  |
| service_neg           | -1.530883755 | 0.050684309    | -30.20429385 | 1.1053E-197 | -1.63022686    | -1.431540649 |
| price_pos             | 0.241605249  | 0.025333677    | 9.536919976  | 1.56966E-21 | 0.191950313    | 0.291260186  |
| price_neg             | -0.608807132 | 0.026651189    | -22.84352646 | 1.3467E-114 | -0.66104444    | -0.556569824 |
| wait_pos              | 0.08109384   | 0.055120568    | 1.471208349  | 0.141244425 | -0.026944497   | 0.189132177  |
| wait_neg              | -0.826013014 | 0.024067792    | -34.32026583 | 1.2865E-253 | -0.873186769   | -0.778839259 |
| cleanliness_pos       | 0.192632798  | 0.032324227    | 5.959393856  | 2.5577E-09  | 0.129276127    | 0.255989469  |
| cleanliness_neg       | -1.278097317 | 0.040735965    | -31.37515759 | 6.5039E-213 | -1.357941303   | -1.198253331 |
| food_pos              | 0.645546765  | 0.01421048     | 45.4275135   | 0           | 0.617693703    | 0.673399827  |
| food_neg              | -0.912905205 | 0.021897724    | -41.68950171 | 0           | -0.955825547   | -0.869984862 |
| value_pos             | 0.179023014  | 0.039602664    | 4.520479101  | 6.19166E-06 | 0.10140034     | 0.256645689  |
| value_neg             | -0.563257129 | 0.087091362    | -6.467428156 | 1.01093E-10 | -0.733959396   | -0.392554862 |
| environment_pos       | 0.262448249  | 0.029506007    | 8.894739527  | 6.1503E-19  | 0.204615392    | 0.320281107  |
| environment_neg       | -0.058072236 | 0.02831554     | -2.050896314 | 0.040285015 | -0.113571733   | -0.002572739 |

## 5. Comprehensive Business Strategy Recommendations

In the end, we conclude restaurant and food business strategy should focus on three main dimensions: food, service, and cleanliness, and integrate them into a comprehensive decision framework based on all the analyses in this project, including entry strategies and improvements strategies.

### ➤ Location selection – *high demand & moderate competition*

In terms of location selection, businesses should prioritize markets with strong demand but not excessive competition, as these areas are more favorable for building repeat visits through consistent food quality.

### ➤ Operations – *avoid failures during peak hours*

When it comes to operations, priority should be given to staffing, service speed, and front- and back-of-house cleanliness during weekend and evening peak hours, because both the check-in analysis and the review regression show that customers are more likely to give low ratings when they experience service failures, long waits, and hygiene issues during peak periods, and the negative effects are significantly stronger than the positive effects. Businesses had better avoid as many failures as possible.

### ➤ Reputation management – *food attracts, service stabilizes, cleanliness protects*

For reputation management, restaurants should leverage high-quality food photos, store presentation, and elite user engagement to strengthen the most important role of food for generating traffic, and also the rating-stabilizing role of service, and the risk-control role of cleanliness.

## Appendices

### [1] Data used overview

-- Null Handling: Null values are treated as 0 in the following analysis using COALESCE unless otherwise specified.  
-- Overview for restaurant businesses statistics: total count, average stars/reviews, geographic coverage and open/closed status

```
SELECT
    COUNT(*) AS total_restaurant_businesses,
    ROUND(AVG(COALESCE(stars, 0)), 2) AS avg_stars,
    ROUND(AVG(COALESCE(review_count, 0)), 2) AS avg_review_count,
    COUNT(DISTINCT city) AS unique_cities,
    COUNT(DISTINCT state) AS unique_states,
    SUM(CASE WHEN is_open = 1 THEN 1 ELSE 0 END) AS open_businesses,
    SUM(CASE WHEN is_open = 0 THEN 1 ELSE 0 END) AS closed_businesses
FROM business
WHERE (categories LIKE '%Restaurant%' OR categories LIKE '%Food%');
```

### [2] Proof of majority data is about food and restaurants

```
SELECT
    CASE
        WHEN categories LIKE '%food%' THEN 'IS FOOD'
        WHEN categories LIKE '%restaurants%' THEN 'IS Restaurants'
        ELSE 'Neither'
    END AS "Category", COUNT(*)
FROM business
GROUP BY Category;
```

### [3] Star rating distribution and average reviews

-- Including business count, percentage of total restaurants, and average review count

```
SELECT
    COALESCE(stars, 0) AS star_rating,
    COUNT(*) AS business_count,
    ROUND(COUNT(*) * 100.0 / (SELECT COUNT(*) FROM business WHERE (categories LIKE '%Restaurant%' OR categories LIKE '%Food%')), 2) AS percentage_of_total,
    ROUND(AVG(COALESCE(review_count, 0)), 2) AS avg_review_count
FROM business
WHERE (categories LIKE '%Restaurant%' OR categories LIKE '%Food%')
GROUP BY COALESCE(stars, 0)
ORDER BY star_rating DESC;
```

### [4] Restaurant demand and competition distribution

```
SELECT
    city,
    state,
    COUNT(*) AS high_rated_business_count,
    ROUND(AVG(COALESCE(stars, 0)), 2) AS avg_star_rating,
    ROUND(AVG(COALESCE(review_count, 0)), 2) AS avg_review_count,
    ROUND(AVG(COALESCE(review_count, 0)) * 1.0 / NULLIF(COUNT(*), 0), 4)
    AS review_business_count_rate
FROM business
WHERE
    (categories LIKE '%Restaurant%' OR categories LIKE '%Food%')
    AND stars >= 4
GROUP BY city, state
HAVING COUNT(*) > 100
ORDER BY avg_review_count DESC;
```

### [5] Monthly review trend

-- Total reviews and average star rating by month

```
SELECT
    strftime('%m', r.date) AS review_month,
    COUNT(*) AS total_reviews,
    ROUND(AVG(COALESCE(r.stars, 0)), 2) AS avg_review_rating
FROM review r
JOIN business b ON r.business_id = b.business_id
WHERE (b.categories LIKE '%Restaurant%' OR b.categories LIKE '%Food%')
GROUP BY review_month
ORDER BY total_reviews DESC;
```

### [6] Service time impact analysis

```
-- Operating hours length vs. star rating/review count
```

```
SELECT
  CASE
    WHEN COALESCE(avg_daily_hours, 0) = 0 THEN '0 Hours (Null/No Hour Data)'
    WHEN COALESCE(avg_daily_hours, 0) < 6 THEN '1-5 Hours'
    WHEN COALESCE(avg_daily_hours, 0) BETWEEN 6 AND 12 THEN '6-12 Hours'
    WHEN COALESCE(avg_daily_hours, 0) BETWEEN 12 AND 18 THEN '12-18 Hours'
    ELSE '≥19 Hours'
  END AS avg_daily_hours_tier,
  COUNT(*) AS business_count,
  ROUND(AVG(COALESCE(b.stars, 0)), 2) AS avg_star_rating,
  ROUND(AVG(COALESCE(b.review_count, 0)), 2) AS avg_review_count
FROM (
  -- Subquery: Extract operating hours from the business table to calculate
  -- average daily operating hours. Cross-midnight cases are handled by adding 24 hours.
  SELECT
    business_id,
    ROUND(
      --Monday operating hours
      (COALESCE(
        (CAST(SUBSTR(b.Monday, INSTR(b.Monday, '-') + 1, 2) AS FLOAT) +
        CAST(SUBSTR(b.Monday, INSTR(b.Monday, '-') + 4, 2) AS FLOAT) / 60) -
        (CAST(SUBSTR(b.Monday, 1, 2) AS FLOAT) + CAST(SUBSTR(b.Monday, 4,
2) AS FLOAT) / 60) +
        CASE
          WHEN
            (CAST(SUBSTR(b.Monday, INSTR(b.Monday, '-') + 1, 2) AS FLOAT) +
            CAST(SUBSTR(b.Monday, INSTR(b.Monday, '-') + 4, 2) AS FLOAT) / 60) <
            (CAST(SUBSTR(b.Monday, 1, 2) AS FLOAT) + CAST(SUBSTR(b.Monday, 4,
2) AS FLOAT) / 60)
          THEN 24 ELSE 0 END, 0
        ) +
      --Tuesday operating hours
      COALESCE(
        (CAST(SUBSTR(b.Tuesday, INSTR(b.Tuesday, '-') + 1, 2) AS FLOAT) +
        CAST(SUBSTR(b.Tuesday, INSTR(b.Tuesday, '-') + 4, 2) AS FLOAT) / 60) -
        (CAST(SUBSTR(b.Tuesday, 1, 2) AS FLOAT) + CAST(SUBSTR(b.Tuesday, 4,
2) AS FLOAT) / 60) +
        CASE
          WHEN
            (CAST(SUBSTR(b.Tuesday, INSTR(b.Tuesday, '-') + 1, 2) AS FLOAT) +
            CAST(SUBSTR(b.Tuesday, INSTR(b.Tuesday, '-') + 4, 2) AS FLOAT) / 60) <
            (CAST(SUBSTR(b.Tuesday, 1, 2) AS FLOAT) + CAST(SUBSTR(b.Tuesday,
4, 2) AS FLOAT) / 60)
          THEN 24 ELSE 0 END, 0
        ) +
      --Wednesday Operating hours
      COALESCE(
        (CAST(SUBSTR(b.Wednesday, INSTR(b.Wednesday, '-') + 1, 2) AS FLOAT) +
        CAST(SUBSTR(b.Wednesday, INSTR(b.Wednesday, '-') + 4, 2) AS FLOAT) / 60) -
        (CAST(SUBSTR(b.Wednesday, 1, 2) AS FLOAT) +
        CAST(SUBSTR(b.Wednesday, 4, 2) AS FLOAT) / 60) +
        CASE
          WHEN
            (CAST(SUBSTR(b.Wednesday, INSTR(b.Wednesday, '-') + 1, 2) AS FLOAT) +
            CAST(SUBSTR(b.Wednesday, INSTR(b.Wednesday, '-') + 4, 2) AS FLOAT) / 60) <
            (CAST(SUBSTR(b.Wednesday, 1, 2) AS FLOAT) +
            CAST(SUBSTR(b.Wednesday, 4, 2) AS FLOAT) / 60)
          THEN 24 ELSE 0 END, 0
        ) +
      --Thursday Operating hours
      COALESCE(
        (CAST(SUBSTR(b.Thursday, INSTR(b.Thursday, '-') + 1, 2) AS FLOAT) +
        CAST(SUBSTR(b.Thursday, INSTR(b.Thursday, '-') + 4, 2) AS FLOAT) / 60) -
        (CAST(SUBSTR(b.Thursday, 1, 2) AS FLOAT) + CAST(SUBSTR(b.Thursday,
4, 2) AS FLOAT) / 60) +
        CASE
          WHEN
            (CAST(SUBSTR(b.Thursday, INSTR(b.Thursday, '-') + 1, 2) AS FLOAT) +
            CAST(SUBSTR(b.Thursday, INSTR(b.Thursday, '-') + 4, 2) AS FLOAT) / 60) <
            (CAST(SUBSTR(b.Thursday, 1, 2) AS FLOAT) + CAST(SUBSTR(b.Thursday,
4, 2) AS FLOAT) / 60)
          THEN 24 ELSE 0 END, 0
        )
    ) AS avg_daily_hours
  )
```

```

        THEN 24 ELSE 0 END, 0
    )+
    --Friday Operating hours
    COALESCE(
        (CAST(SUBSTR(b.Friday, INSTR(b.Friday, '-')+1, 2) AS FLOAT) +
        CAST(SUBSTR(b.Friday, INSTR(b.Friday, '-')+4, 2) AS FLOAT)/60) -
        (CAST(SUBSTR(b.Friday, 1, 2) AS FLOAT) + CAST(SUBSTR(b.Friday, 4,
2) AS FLOAT)/60) +
        CASE
        WHEN
            (CAST(SUBSTR(b.Friday, INSTR(b.Friday, '-')+1, 2) AS FLOAT) +
            CAST(SUBSTR(b.Friday, INSTR(b.Friday, '-')+4, 2) AS FLOAT)/60)<
            (CAST(SUBSTR(b.Friday, 1, 2) AS FLOAT) + CAST(SUBSTR(b.Friday, 4,
2) AS FLOAT)/60)
        THEN 24 ELSE 0 END, 0
    ))/5

    )AS avg_daily_hours
FROM business b
    ) h
LEFT JOIN business b ON h.business_id = b.business_id
WHERE (b.categories LIKE '%Restaurant%' OR b.categories LIKE '%Food%')
GROUP BY avg_daily_hours_tier
ORDER BY avg_daily_hours;

```

### [7] Photo count analysis

```

-- User photo count vs. star rating/review count
SELECT
    CASE
        WHEN COALESCE(total_photos, 0) = 0 THEN '0 Photos (Null/No Photos)'
        WHEN COALESCE(total_photos, 0) < 10 THEN '1-9 Photos'
        WHEN COALESCE(total_photos, 0) BETWEEN 10 AND 49 THEN '10-49 Photos'
        WHEN COALESCE(total_photos, 0) BETWEEN 50 AND 99 THEN '50-99 Photos'
        ELSE '≥100 Photos'
    END AS user_photo_count_tier,
    COUNT(*) AS business_count,
    ROUND(AVG(COALESCE(b.stars, 0)), 2) AS avg_star_rating,
    ROUND(AVG(COALESCE(b.review_count, 0)), 2) AS avg_review_count
FROM (
    -- Subquery: Count total photos and label-specific photos
    SELECT
        business_id,
        COUNT(*) AS total_photos,
        SUM(CASE WHEN COALESCE(label, 'Unknown') LIKE '%Food%' THEN 1 ELSE 0 END)
AS food_photo_count,
        SUM(CASE WHEN COALESCE(label, 'Unknown') LIKE '%Interior%' THEN 1 ELSE 0
END) AS interior_photo_count,
        SUM(CASE WHEN COALESCE(label, 'Unknown') LIKE '%Service%' THEN 1 ELSE 0
END) AS service_photo_count
    FROM photos
    GROUP BY business_id
) p
LEFT JOIN business b ON p.business_id = b.business_id --Left join to retain all
restaurants (including no photo data)
WHERE (b.categories LIKE '%Restaurant%' OR b.categories LIKE '%Food%')
GROUP BY user_photo_count_tier
ORDER BY total_photos;

```

### [8] Photo label impact analysis

```

SELECT
    p.label AS photo_label,
    COUNT(*) AS photo_count,
    ROUND(AVG(b.stars), 2) AS avg_business_stars
FROM photos p
JOIN business b ON p.business_id = b.business_id
WHERE (b.categories LIKE '%Restaurant%' OR b.categories LIKE '%Food%')
GROUP BY photo_label
ORDER BY avg_business_stars DESC;

```

### [9] Photo label combo impact analysis

```

-- One label
SELECT

```



```

1 AS combo_size,
p.label AS label1,
'' AS label2,
'' AS label3,
COUNT(DISTINCT b.business_id) AS business_count,
ROUND(AVG(b.stars), 2) AS avg_stars
FROM photos p
JOIN business b ON p.business_id = b.business_id
WHERE (b.categories LIKE '%Restaurant%' OR b.categories LIKE '%Food%')
AND p.label IS NOT NULL
GROUP BY p.label

UNION ALL

-- Two labels
SELECT
2 AS combo_size,
a.label AS label1,
b.label AS label2,
'' AS label3,
COUNT(DISTINCT p1.business_id) AS business_count,
ROUND(AVG(b2.stars), 2) AS avg_stars
FROM (SELECT DISTINCT label FROM photos WHERE label IS NOT NULL LIMIT 5) a
JOIN (SELECT DISTINCT label FROM photos WHERE label IS NOT NULL LIMIT 5) b ON
a.label < b.label
JOIN photos p1 ON p1.label = a.label
JOIN photos p2 ON p2.business_id = p1.business_id AND p2.label = b.label
JOIN business b2 ON p1.business_id = b2.business_id
WHERE b2.categories LIKE '%Restaurant%' OR b2.categories LIKE '%Food%'
GROUP BY a.label, b.label

UNION ALL

-- Three labels
SELECT
3 AS combo_size,
a.label AS label1,
b.label AS label2,
c.label AS label3,
COUNT(DISTINCT p1.business_id) AS business_count,
ROUND(AVG(b2.stars), 2) AS avg_stars
FROM (SELECT DISTINCT label FROM photos WHERE label IS NOT NULL LIMIT 5) a
JOIN (SELECT DISTINCT label FROM photos WHERE label IS NOT NULL LIMIT 5) b ON
a.label < b.label
JOIN (SELECT DISTINCT label FROM photos WHERE label IS NOT NULL LIMIT 5) c ON
b.label < c.label
JOIN photos p1 ON p1.label = a.label
JOIN photos p2 ON p2.business_id = p1.business_id AND p2.label = b.label
JOIN photos p3 ON p3.business_id = p1.business_id AND p3.label = c.label
JOIN business b2 ON p1.business_id = b2.business_id
WHERE b2.categories LIKE '%Restaurant%' OR b2.categories LIKE '%Food%'
GROUP BY a.label, b.label, c.label

ORDER BY combo size ASC, business count DESC;

```

## [10] User behavior trend

```

-- Elite vs. regular user review behavior by restaurant star rating
SELECT
COALESCE(b.stars, 0) AS restaurant_star_rating,
CASE WHEN u.elite IS NOT NULL AND u.elite != '' THEN 'Elite User' ELSE 'Regular
User' END AS user_type,
COUNT(*) AS review_count,
ROUND(AVG(COALESCE(r.stars, 0)), 2) AS avg_review_rating
FROM review r
JOIN business b ON r.business_id = b.business_id
JOIN user u ON r.user_id = u.user_id
WHERE (b.categories LIKE '%Restaurant%' OR b.categories LIKE '%Food%')
GROUP BY
COALESCE(b.stars, 0),
CASE WHEN u.elite IS NOT NULL AND u.elite != '' THEN 'Elite User' ELSE 'Regular
User' END
ORDER BY

```

```
COALESCE(b.stars, 0) DESC,
user_type DESC;
```

#### [11] User statistics summary

```
-- total users, elite user count/ratio and average reviews/stars/fans
SELECT
    COUNT(DISTINCT u.user_id) AS total_restaurant_reviewers,
    --Count the number of elite users
    COUNT(DISTINCT CASE WHEN u.elite IS NOT NULL AND u.elite != '' THEN u.user_id
END) AS elite_user_count,
    --Calculate the percentage of elite users
    ROUND(
        COUNT(DISTINCT CASE WHEN u.elite IS NOT NULL AND u.elite != '' THEN
u.user_id END) * 100.0 /
        NULLIF(COUNT(DISTINCT u.user_id), 0),
        2
    ) AS elite_user_percentage,
    ROUND(AVG(COALESCE(u.review_count, 0)), 2) AS avg_user_total_reviews,
    ROUND(AVG(COALESCE(u.average_stars, 0)), 2) AS avg_user_rating,
    ROUND(AVG(COALESCE(u.fans, 0)), 2) AS avg_user_fans
FROM user u
JOIN review r ON u.user_id = r.user_id
JOIN business b ON r.business_id = b.business_id

WHERE (b.categories LIKE '%Restaurant%' OR b.categories LIKE '%Food%');
```

#### [12] Check-in count analysis

```
SELECT
    CASE WHEN ch.business_id IS NULL THEN '0 Check-Ins' ELSE 'Has Check-Ins' END AS
checkin_status,
    COUNT(DISTINCT b.business_id) AS business_count,
    ROUND(AVG(b.stars), 2) AS avg_stars,
    ROUND(AVG(b.review_count), 2) AS avg_reviews
FROM business b
LEFT JOIN checkin ch ON b.business_id = ch.business_id
WHERE (b.categories LIKE '%Restaurant%' OR b.categories LIKE '%Food%')
GROUP BY checkin_status
ORDER BY avg_stars DESC;
```

#### [13] Weekly distribution of check-in activity

```
WITH filtered_checkin AS (
    SELECT ch.*
    FROM checkin ch
    JOIN business b ON b.business_id = ch.business_id
    WHERE (b.categories LIKE '%Restaurant%' OR b.categories LIKE '%Food%')),
    unpivot AS (
        SELECT 'time.Thursday.21:00' AS col_name, [time.Thursday.21:00] AS val
    FROM filtered_checkin
    UNION ALL SELECT 'time.Thursday.1:00', [time.Thursday.1:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Thursday.4:00', [time.Thursday.4:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Thursday.2:00', [time.Thursday.2:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Thursday.20:00', [time.Thursday.20:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Thursday.22:00', [time.Thursday.22:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Thursday.19:00', [time.Thursday.19:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Thursday.18:00', [time.Thursday.18:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Thursday.15:00', [time.Thursday.15:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Thursday.13:00', [time.Thursday.13:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Thursday.23:00', [time.Thursday.23:00] FROM
filtered_checkin

    UNION ALL SELECT 'time.Wednesday.11:00', [time.Wednesday.11:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Wednesday.13:00', [time.Wednesday.13:00] FROM
```

```

filtered_checkin
    UNION ALL SELECT 'time.Wednesday.14:00', [time.Wednesday.14:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Wednesday.17:00', [time.Wednesday.17:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Wednesday.6:00', [time.Wednesday.6:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Wednesday.2:00', [time.Wednesday.2:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Wednesday.0:00', [time.Wednesday.0:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Wednesday.1:00', [time.Wednesday.1:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Wednesday.21:00', [time.Wednesday.21:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Wednesday.18:00', [time.Wednesday.18:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Wednesday.19:00', [time.Wednesday.19:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Wednesday.20:00', [time.Wednesday.20:00] FROM
filtered_checkin

    UNION ALL SELECT 'time.Sunday.18:00', [time.Sunday.18:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Sunday.16:00', [time.Sunday.16:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Sunday.14:00', [time.Sunday.14:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Sunday.19:00', [time.Sunday.19:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Sunday.17:00', [time.Sunday.17:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Sunday.23:00', [time.Sunday.23:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Sunday.21:00', [time.Sunday.21:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Sunday.20:00', [time.Sunday.20:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Sunday.6:00', [time.Sunday.6:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Sunday.0:00', [time.Sunday.0:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Sunday.2:00', [time.Sunday.2:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Sunday.3:00', [time.Sunday.3:00] FROM
filtered_checkin

    UNION ALL SELECT 'time.Friday.16:00', [time.Friday.16:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Friday.14:00', [time.Friday.14:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Friday.10:00', [time.Friday.10:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Friday.23:00', [time.Friday.23:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Friday.18:00', [time.Friday.18:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Friday.17:00', [time.Friday.17:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Friday.21:00', [time.Friday.21:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Friday.19:00', [time.Friday.19:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Friday.15:00', [time.Friday.15:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Friday.22:00', [time.Friday.22:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Friday.3:00', [time.Friday.3:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Friday.0:00', [time.Friday.0:00] FROM
filtered_checkin

```

```

        UNION ALL SELECT 'time.Saturday.21:00', [time.Saturday.21:00] FROM
filtered_checkin
        UNION ALL SELECT 'time.Saturday.23:00', [time.Saturday.23:00] FROM
filtered_checkin
        UNION ALL SELECT 'time.Saturday.18:00', [time.Saturday.18:00] FROM
filtered_checkin
        UNION ALL SELECT 'time.Saturday.10:00', [time.Saturday.10:00] FROM
filtered_checkin
        UNION ALL SELECT 'time.Saturday.12:00', [time.Saturday.12:00] FROM
filtered_checkin
        UNION ALL SELECT 'time.Saturday.13:00', [time.Saturday.13:00] FROM
filtered_checkin
        UNION ALL SELECT 'time.Saturday.15:00', [time.Saturday.15:00] FROM
filtered_checkin
        UNION ALL SELECT 'time.Saturday.16:00', [time.Saturday.16:00] FROM
filtered_checkin
        UNION ALL SELECT 'time.Saturday.17:00', [time.Saturday.17:00] FROM
filtered_checkin
        UNION ALL SELECT 'time.Saturday.14:00', [time.Saturday.14:00] FROM
filtered_checkin
        UNION ALL SELECT 'time.Saturday.2:00', [time.Saturday.2:00] FROM
filtered_checkin
        UNION ALL SELECT 'time.Saturday.0:00', [time.Saturday.0:00] FROM
filtered_checkin
        UNION ALL SELECT 'time.Saturday.1:00', [time.Saturday.1:00] FROM
filtered_checkin

        UNION ALL SELECT 'time.Monday.12:00', [time.Monday.12:00] FROM
filtered_checkin
        UNION ALL SELECT 'time.Monday.11:00', [time.Monday.11:00] FROM
filtered_checkin
        UNION ALL SELECT 'time.Monday.17:00', [time.Monday.17:00] FROM
filtered_checkin
        UNION ALL SELECT 'time.Monday.14:00', [time.Monday.14:00] FROM
filtered_checkin
        UNION ALL SELECT 'time.Monday.15:00', [time.Monday.15:00] FROM
filtered_checkin
        UNION ALL SELECT 'time.Monday.18:00', [time.Monday.18:00] FROM
filtered_checkin
        UNION ALL SELECT 'time.Monday.4:00', [time.Monday.4:00] FROM
filtered_checkin
        UNION ALL SELECT 'time.Monday.23:00', [time.Monday.23:00] FROM
filtered_checkin
        UNION ALL SELECT 'time.Monday.20:00', [time.Monday.20:00] FROM
filtered_checkin
        UNION ALL SELECT 'time.Monday.19:00', [time.Monday.19:00] FROM
filtered_checkin

        UNION ALL SELECT 'time.Tuesday.18:00', [time.Tuesday.18:00] FROM
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        UNION ALL SELECT 'time.Tuesday.13:00', [time.Tuesday.13:00] FROM
filtered_checkin
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filtered_checkin
        UNION ALL SELECT 'time.Tuesday.17:00', [time.Tuesday.17:00] FROM
filtered_checkin
        UNION ALL SELECT 'time.Tuesday.15:00', [time.Tuesday.15:00] FROM
filtered_checkin
        UNION ALL SELECT 'time.Tuesday.4:00', [time.Tuesday.4:00] FROM
filtered_checkin
        UNION ALL SELECT 'time.Tuesday.21:00', [time.Tuesday.21:00] FROM
filtered_checkin
        UNION ALL SELECT 'time.Tuesday.20:00', [time.Tuesday.20:00] FROM
filtered_checkin
        UNION ALL SELECT 'time.Tuesday.23:00', [time.Tuesday.23:00] FROM
filtered_checkin
    )
SELECT
    substr(col_name, 6, instr(substr(col_name, 6), '.') - 1) AS weekday,
    COALESCE(SUM(CAST(NULLIF(val, 'N/A') AS INTEGER)), 0) AS weekday sum

```



```
FROM unpivot
GROUP BY weekday
ORDER BY weekday sum DESC;
```

#### [14] Hourly distribution of check-in activity

```
WITH filtered_checkin AS (
    SELECT ch.*
    FROM checkin ch
    JOIN business b ON b.business_id = ch.business_id
    WHERE (b.categories LIKE '%Restaurant%' OR b.categories LIKE '%Food%')),
    unpivot AS (
        SELECT 'time.Thursday.21:00' AS col_name, [time.Thursday.21:00] AS val
    FROM filtered_checkin
        UNION ALL SELECT 'time.Thursday.1:00', [time.Thursday.1:00] FROM
    filtered_checkin
        UNION ALL SELECT 'time.Thursday.4:00', [time.Thursday.4:00] FROM
    filtered_checkin
        UNION ALL SELECT 'time.Thursday.2:00', [time.Thursday.2:00] FROM
    filtered_checkin
        UNION ALL SELECT 'time.Thursday.20:00', [time.Thursday.20:00] FROM
    filtered_checkin
        UNION ALL SELECT 'time.Thursday.22:00', [time.Thursday.22:00] FROM
    filtered_checkin
        UNION ALL SELECT 'time.Thursday.19:00', [time.Thursday.19:00] FROM
    filtered_checkin
        UNION ALL SELECT 'time.Thursday.18:00', [time.Thursday.18:00] FROM
    filtered_checkin
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    filtered_checkin
        UNION ALL SELECT 'time.Thursday.13:00', [time.Thursday.13:00] FROM
    filtered_checkin
        UNION ALL SELECT 'time.Thursday.23:00', [time.Thursday.23:00] FROM
    filtered_checkin

        UNION ALL SELECT 'time.Wednesday.11:00', [time.Wednesday.11:00] FROM
    filtered_checkin
        UNION ALL SELECT 'time.Wednesday.13:00', [time.Wednesday.13:00] FROM
    filtered_checkin
        UNION ALL SELECT 'time.Wednesday.14:00', [time.Wednesday.14:00] FROM
    filtered_checkin
        UNION ALL SELECT 'time.Wednesday.17:00', [time.Wednesday.17:00] FROM
    filtered_checkin
        UNION ALL SELECT 'time.Wednesday.6:00', [time.Wednesday.6:00] FROM
    filtered_checkin
        UNION ALL SELECT 'time.Wednesday.2:00', [time.Wednesday.2:00] FROM
    filtered_checkin
        UNION ALL SELECT 'time.Wednesday.0:00', [time.Wednesday.0:00] FROM
    filtered_checkin
        UNION ALL SELECT 'time.Wednesday.1:00', [time.Wednesday.1:00] FROM
    filtered_checkin
        UNION ALL SELECT 'time.Wednesday.21:00', [time.Wednesday.21:00] FROM
    filtered_checkin
        UNION ALL SELECT 'time.Wednesday.18:00', [time.Wednesday.18:00] FROM
    filtered_checkin
        UNION ALL SELECT 'time.Wednesday.19:00', [time.Wednesday.19:00] FROM
    filtered_checkin
        UNION ALL SELECT 'time.Wednesday.20:00', [time.Wednesday.20:00] FROM
    filtered_checkin

        UNION ALL SELECT 'time.Sunday.18:00', [time.Sunday.18:00] FROM
    filtered_checkin
        UNION ALL SELECT 'time.Sunday.16:00', [time.Sunday.16:00] FROM
    filtered_checkin
        UNION ALL SELECT 'time.Sunday.14:00', [time.Sunday.14:00] FROM
    filtered_checkin
        UNION ALL SELECT 'time.Sunday.19:00', [time.Sunday.19:00] FROM
    filtered_checkin
        UNION ALL SELECT 'time.Sunday.17:00', [time.Sunday.17:00] FROM
    filtered_checkin
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    filtered_checkin
        UNION ALL SELECT 'time.Sunday.21:00', [time.Sunday.21:00] FROM
    filtered_checkin
```

```

filtered_checkin
    UNION ALL SELECT 'time.Sunday.20:00', [time.Sunday.20:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Sunday.6:00', [time.Sunday.6:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Sunday.0:00', [time.Sunday.0:00] FROM
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    UNION ALL SELECT 'time.Sunday.2:00', [time.Sunday.2:00] FROM
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    UNION ALL SELECT 'time.Friday.16:00', [time.Friday.16:00] FROM
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    UNION ALL SELECT 'time.Friday.22:00', [time.Friday.22:00] FROM
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    UNION ALL SELECT 'time.Friday.3:00', [time.Friday.3:00] FROM
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    UNION ALL SELECT 'time.Friday.0:00', [time.Friday.0:00] FROM
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    UNION ALL SELECT 'time.Saturday.21:00', [time.Saturday.21:00] FROM
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    UNION ALL SELECT 'time.Saturday.23:00', [time.Saturday.23:00] FROM
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    UNION ALL SELECT 'time.Saturday.18:00', [time.Saturday.18:00] FROM
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    UNION ALL SELECT 'time.Saturday.10:00', [time.Saturday.10:00] FROM
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    UNION ALL SELECT 'time.Saturday.16:00', [time.Saturday.16:00] FROM
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    UNION ALL SELECT 'time.Saturday.17:00', [time.Saturday.17:00] FROM
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    UNION ALL SELECT 'time.Saturday.14:00', [time.Saturday.14:00] FROM
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    UNION ALL SELECT 'time.Saturday.2:00', [time.Saturday.2:00] FROM
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    UNION ALL SELECT 'time.Saturday.0:00', [time.Saturday.0:00] FROM
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    UNION ALL SELECT 'time.Saturday.1:00', [time.Saturday.1:00] FROM
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    UNION ALL SELECT 'time.Monday.12:00', [time.Monday.12:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Monday.11:00', [time.Monday.11:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Monday.17:00', [time.Monday.17:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Monday.14:00', [time.Monday.14:00] FROM
filtered_checkin

```

```

        UNION ALL SELECT 'time.Monday.15:00', [time.Monday.15:00] FROM
filtered_checkin
        UNION ALL SELECT 'time.Monday.18:00', [time.Monday.18:00] FROM
filtered_checkin
        UNION ALL SELECT 'time.Monday.4:00', [time.Monday.4:00] FROM
filtered_checkin
        UNION ALL SELECT 'time.Monday.23:00', [time.Monday.23:00] FROM
filtered_checkin
        UNION ALL SELECT 'time.Monday.20:00', [time.Monday.20:00] FROM
filtered_checkin
        UNION ALL SELECT 'time.Monday.19:00', [time.Monday.19:00] FROM
filtered_checkin

        UNION ALL SELECT 'time.Tuesday.18:00', [time.Tuesday.18:00] FROM
filtered_checkin
        UNION ALL SELECT 'time.Tuesday.12:00', [time.Tuesday.12:00] FROM
filtered_checkin
        UNION ALL SELECT 'time.Tuesday.13:00', [time.Tuesday.13:00] FROM
filtered_checkin
        UNION ALL SELECT 'time.Tuesday.16:00', [time.Tuesday.16:00] FROM
filtered_checkin
        UNION ALL SELECT 'time.Tuesday.17:00', [time.Tuesday.17:00] FROM
filtered_checkin
        UNION ALL SELECT 'time.Tuesday.15:00', [time.Tuesday.15:00] FROM
filtered_checkin
        UNION ALL SELECT 'time.Tuesday.4:00', [time.Tuesday.4:00] FROM
filtered_checkin
        UNION ALL SELECT 'time.Tuesday.21:00', [time.Tuesday.21:00] FROM
filtered_checkin
        UNION ALL SELECT 'time.Tuesday.20:00', [time.Tuesday.20:00] FROM
filtered_checkin
        UNION ALL SELECT 'time.Tuesday.23:00', [time.Tuesday.23:00] FROM
filtered_checkin
    ),
    hour_agg AS (
        SELECT
            substr(col_name, 6 + instr(substr(col_name, 6), '.')) AS hour,
            COALESCE(SUM(CAST(NULLIF(val, 'N/A') AS INTEGER)), 0) AS hour_sum
        FROM unpivot
        GROUP BY hour
    )
    SELECT hour, hour_sum
    FROM hour_agg
    ORDER BY hour sum DESC;

```

#### [15] Day-of-week analysis of check-in activity

```

WITH filtered_checkin AS (
    SELECT ch.*
    FROM checkin ch
    JOIN business b ON b.business_id = ch.business_id
    WHERE (b.categories LIKE '%Restaurant%' OR b.categories LIKE '%Food%')),
    unpivot AS (
        SELECT 'time.Thursday.21:00' AS col_name, [time.Thursday.21:00] AS val
    FROM filtered_checkin
        UNION ALL SELECT 'time.Thursday.1:00', [time.Thursday.1:00] FROM
filtered_checkin
        UNION ALL SELECT 'time.Thursday.4:00', [time.Thursday.4:00] FROM
filtered_checkin
        UNION ALL SELECT 'time.Thursday.2:00', [time.Thursday.2:00] FROM
filtered_checkin
        UNION ALL SELECT 'time.Thursday.20:00', [time.Thursday.20:00] FROM
filtered_checkin
        UNION ALL SELECT 'time.Thursday.22:00', [time.Thursday.22:00] FROM
filtered_checkin
        UNION ALL SELECT 'time.Thursday.19:00', [time.Thursday.19:00] FROM
filtered_checkin
        UNION ALL SELECT 'time.Thursday.18:00', [time.Thursday.18:00] FROM
filtered_checkin
        UNION ALL SELECT 'time.Thursday.15:00', [time.Thursday.15:00] FROM
filtered_checkin
        UNION ALL SELECT 'time.Thursday.13:00', [time.Thursday.13:00] FROM
filtered_checkin

```

```

UNION ALL SELECT 'time.Thursday.23:00', [time.Thursday.23:00] FROM
filtered_checkin

```

```

UNION ALL SELECT 'time.Wednesday.11:00', [time.Wednesday.11:00] FROM
filtered_checkin

```

```

UNION ALL SELECT 'time.Wednesday.13:00', [time.Wednesday.13:00] FROM
filtered_checkin

```

```

UNION ALL SELECT 'time.Wednesday.14:00', [time.Wednesday.14:00] FROM
filtered_checkin

```

```

UNION ALL SELECT 'time.Wednesday.17:00', [time.Wednesday.17:00] FROM
filtered_checkin

```

```

UNION ALL SELECT 'time.Wednesday.6:00', [time.Wednesday.6:00] FROM
filtered_checkin

```

```

UNION ALL SELECT 'time.Wednesday.2:00', [time.Wednesday.2:00] FROM
filtered_checkin

```

```

UNION ALL SELECT 'time.Wednesday.0:00', [time.Wednesday.0:00] FROM
filtered_checkin

```

```

UNION ALL SELECT 'time.Wednesday.1:00', [time.Wednesday.1:00] FROM
filtered_checkin

```

```

UNION ALL SELECT 'time.Wednesday.21:00', [time.Wednesday.21:00] FROM
filtered_checkin

```

```

UNION ALL SELECT 'time.Wednesday.18:00', [time.Wednesday.18:00] FROM
filtered_checkin

```

```

UNION ALL SELECT 'time.Wednesday.19:00', [time.Wednesday.19:00] FROM
filtered_checkin

```

```

UNION ALL SELECT 'time.Wednesday.20:00', [time.Wednesday.20:00] FROM
filtered_checkin

```

```

UNION ALL SELECT 'time.Sunday.18:00', [time.Sunday.18:00] FROM
filtered_checkin

```

```

UNION ALL SELECT 'time.Sunday.16:00', [time.Sunday.16:00] FROM
filtered_checkin

```

```

UNION ALL SELECT 'time.Sunday.14:00', [time.Sunday.14:00] FROM
filtered_checkin

```

```

UNION ALL SELECT 'time.Sunday.19:00', [time.Sunday.19:00] FROM
filtered_checkin

```

```

UNION ALL SELECT 'time.Sunday.17:00', [time.Sunday.17:00] FROM
filtered_checkin

```

```

UNION ALL SELECT 'time.Sunday.23:00', [time.Sunday.23:00] FROM
filtered_checkin

```

```

UNION ALL SELECT 'time.Sunday.21:00', [time.Sunday.21:00] FROM
filtered_checkin

```

```

UNION ALL SELECT 'time.Sunday.20:00', [time.Sunday.20:00] FROM
filtered_checkin

```

```

UNION ALL SELECT 'time.Sunday.6:00', [time.Sunday.6:00] FROM
filtered_checkin

```

```

UNION ALL SELECT 'time.Sunday.0:00', [time.Sunday.0:00] FROM
filtered_checkin

```

```

UNION ALL SELECT 'time.Sunday.2:00', [time.Sunday.2:00] FROM
filtered_checkin

```

```

UNION ALL SELECT 'time.Sunday.3:00', [time.Sunday.3:00] FROM
filtered_checkin

```

```

UNION ALL SELECT 'time.Friday.16:00', [time.Friday.16:00] FROM
filtered_checkin

```

```

UNION ALL SELECT 'time.Friday.14:00', [time.Friday.14:00] FROM
filtered_checkin

```

```

UNION ALL SELECT 'time.Friday.10:00', [time.Friday.10:00] FROM
filtered_checkin

```

```

UNION ALL SELECT 'time.Friday.23:00', [time.Friday.23:00] FROM
filtered_checkin

```

```

UNION ALL SELECT 'time.Friday.18:00', [time.Friday.18:00] FROM
filtered_checkin

```

```

UNION ALL SELECT 'time.Friday.17:00', [time.Friday.17:00] FROM
filtered_checkin

```

```

UNION ALL SELECT 'time.Friday.21:00', [time.Friday.21:00] FROM
filtered_checkin

```

```

UNION ALL SELECT 'time.Friday.19:00', [time.Friday.19:00] FROM
filtered_checkin

```

```

UNION ALL SELECT 'time.Friday.15:00', [time.Friday.15:00] FROM
filtered_checkin

```

```

UNION ALL SELECT 'time.Friday.22:00', [time.Friday.22:00] FROM

```



```
filtered_checkin
    UNION ALL SELECT 'time.Friday.3:00', [time.Friday.3:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Friday.0:00', [time.Friday.0:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Saturday.21:00', [time.Saturday.21:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Saturday.23:00', [time.Saturday.23:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Saturday.18:00', [time.Saturday.18:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Saturday.10:00', [time.Saturday.10:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Saturday.12:00', [time.Saturday.12:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Saturday.13:00', [time.Saturday.13:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Saturday.15:00', [time.Saturday.15:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Saturday.16:00', [time.Saturday.16:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Saturday.17:00', [time.Saturday.17:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Saturday.14:00', [time.Saturday.14:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Saturday.2:00', [time.Saturday.2:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Saturday.0:00', [time.Saturday.0:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Saturday.1:00', [time.Saturday.1:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Monday.12:00', [time.Monday.12:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Monday.11:00', [time.Monday.11:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Monday.17:00', [time.Monday.17:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Monday.14:00', [time.Monday.14:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Monday.15:00', [time.Monday.15:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Monday.18:00', [time.Monday.18:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Monday.4:00', [time.Monday.4:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Monday.23:00', [time.Monday.23:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Monday.20:00', [time.Monday.20:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Monday.19:00', [time.Monday.19:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Tuesday.18:00', [time.Tuesday.18:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Tuesday.12:00', [time.Tuesday.12:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Tuesday.13:00', [time.Tuesday.13:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Tuesday.16:00', [time.Tuesday.16:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Tuesday.17:00', [time.Tuesday.17:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Tuesday.15:00', [time.Tuesday.15:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Tuesday.4:00', [time.Tuesday.4:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Tuesday.21:00', [time.Tuesday.21:00] FROM
filtered_checkin
    UNION ALL SELECT 'time.Tuesday.20:00', [time.Tuesday.20:00] FROM
filtered_checkin
```



```

        UNION ALL SELECT 'time.Tuesday.23:00', [time.Tuesday.23:00] FROM
filtered_checkin
),
agg AS (
    SELECT
        substr(col_name, 6, instr(substr(col_name, 6), '.')) - 1 AS weekday,
        substr(col_name, 6 + instr(substr(col_name, 6), '.')) AS hour,
        SUM(CAST(NULLIF(val, 'N/A') AS INTEGER)) AS sum_val
    FROM unpivot
    GROUP BY weekday, hour
)
SELECT weekday, hour, sum_val
FROM agg
ORDER BY
    CASE weekday
        WHEN 'Monday' THEN 1 WHEN 'Tuesday' THEN 2 WHEN 'Wednesday' THEN 3
        WHEN 'Thursday' THEN 4 WHEN 'Friday' THEN 5 WHEN 'Saturday' THEN 6
        WHEN 'Sunday' THEN 7 ELSE 99 END,
    sum_val DESC;

```

#### [16] Review summarization

```

SELECT
    CASE
        WHEN COALESCE(review_count, 0) = 0 THEN '0'
        WHEN COALESCE(review_count, 0) BETWEEN 1 AND 9 THEN '1-9'
        WHEN COALESCE(review_count, 0) BETWEEN 10 AND 49 THEN '10-49'
        WHEN COALESCE(review_count, 0) BETWEEN 50 AND 100 THEN '50-100'
        WHEN COALESCE(review_count, 0) BETWEEN 101 AND 199 THEN '100-199'
        WHEN COALESCE(review_count, 0) BETWEEN 200 AND 299 THEN '200-299'
        WHEN COALESCE(review_count, 0) BETWEEN 300 AND 399 THEN '300-399'
        WHEN COALESCE(review_count, 0) BETWEEN 400 AND 499 THEN '400-499'
        ELSE '500+'
    END AS review_count_tier,
    COUNT(*) AS business_count,
    ROUND(AVG(COALESCE(stars, 0)), 2) AS avg_star_rating,
    ROUND(AVG(COALESCE(review_count, 0)), 2) AS avg_review_count
FROM business
WHERE (categories LIKE '%Restaurant%' OR categories LIKE '%Food%')
GROUP BY review_count_tier
ORDER BY
    CASE review_count_tier
        WHEN '0' THEN 1
        WHEN '1-9' THEN 2
        WHEN '10-49' THEN 3
        WHEN '50-100' THEN 4
        WHEN '100-199' THEN 5
        WHEN '200-299' THEN 6
        WHEN '300-399' THEN 7
        WHEN '400-499' THEN 8
        WHEN '500+' THEN 9
    END;

```

#### [17] Positive and negative words in review

```

CREATE TABLE review_tags AS
SELECT review_id
    , business_id
    , stars
    , DATE
    , TEXT
    , Year
    , service_pos
    , service_neg
    , price_pos
    , price_neg
    , wait_pos
    , wait_neg
    , cleanliness_pos
    , cleanliness_neg
    , food_pos
    , food_neg
    , value_pos
    , value_neg

```

```

,environment_pos
,environment_neg
FROM (
    SELECT review_id
           ,business_id
           ,stars
           ,DATE
           ,TEXT
           ,Year
           /
           -- Service positive
           CASE
               WHEN t LIKE '% good service %'
                   OR t LIKE '% great service %'
                   OR t LIKE '% excellent service %'
                   OR t LIKE '% amazing service %'
                   OR t LIKE '% friendly service %'
                   OR t LIKE '% friendly staff %'
                   OR t LIKE '% friendly server %'
                   OR t LIKE '% friendly waiter %'
                   OR t LIKE '% friendly waitress %'
                   OR t LIKE '% helpful staff %'
                   OR t LIKE '% attentive service %'
                   OR t LIKE '% attentive staff %'
                   OR t LIKE '% professional service %'
                   OR t LIKE '% prompt service %'
                   OR t LIKE '% quick service %'
                   OR t LIKE '% polite %'
                   OR t LIKE '% courteous %'
                   OR t LIKE '% great staff %'
                   OR t LIKE '% excellent staff %'
               THEN 1
               ELSE 0
           END AS service_pos
           /
           -- Service negative
           CASE
               WHEN t LIKE '% bad service %'
                   OR t LIKE '% poor service %'
                   OR t LIKE '% terrible service %'
                   OR t LIKE '% awful service %'
                   OR t LIKE '% horrible service %'
                   OR t LIKE '% slow service %'
                   OR t LIKE '% rude service %'
                   OR t LIKE '% rude staff %'
                   OR t LIKE '% unfriendly staff %'
                   OR t LIKE '% unprofessional service %'
                   OR t LIKE '% unprofessional staff %'
                   OR t LIKE '% ignored us %'
                   OR t LIKE '% ignored me %'
                   OR t LIKE '% inattentive service %'
                   OR t LIKE '% dismissive staff %'
                   OR t LIKE '% server was rude %'
                   OR t LIKE '% waiter was rude %'
                   OR t LIKE '% waitress was rude %'
                   OR t LIKE '% host was rude %'
                   OR t LIKE '% hostess was rude %'
                   OR t LIKE '% manager was rude %'
               THEN 1
               ELSE 0
           END AS service_neg
           /
           -- Price positive
           CASE
               WHEN t LIKE '% affordable %'
                   OR t LIKE '% reasonably priced %'
                   OR t LIKE '% reasonable prices %'
                   OR t LIKE '% reasonable price %'
                   OR t LIKE '% cheap %'
                   OR t LIKE '% inexpensive %'
                   OR t LIKE '% fair price %'
                   OR t LIKE '% good value %'

```

```

        OR t LIKE '% great value %'
        OR t LIKE '% worth it %'
        OR t LIKE '% worth every penny %'
        OR t LIKE '% value for money %'
        THEN 1
    ELSE 0
    END AS price_pos
/
-- Price negative
CASE
    WHEN t LIKE '% expensive %'
        OR t LIKE '% overpriced %'
        OR t LIKE '% pricey %'
        OR t LIKE '% not worth %'
        OR t LIKE '% not worth it %'
        OR t LIKE '% rip off %'
        OR t LIKE '% ripoff %'
        OR t LIKE '% overcharged %'
        OR t LIKE '% over charged %'
        OR t LIKE '% charged extra %'
        OR t LIKE '% hidden fee %'
        OR t LIKE '% hidden fees %'
        OR t LIKE '% gratuity included %'
        OR t LIKE '% automatic gratuity %'
        OR t LIKE '% auto gratuity %'
        OR t LIKE '% refund %'
        OR t LIKE '% wrong bill %'
        OR t LIKE '% incorrect bill %'
        THEN 1
    ELSE 0
    END AS price_neg
/
-- Wait positive
CASE
    WHEN t LIKE '% no wait %'
        OR t LIKE '% no line %'
        OR t LIKE '% seated right away %'
        OR t LIKE '% sat right away %'
        OR t LIKE '% prompt service %'
        OR t LIKE '% quick service %'
        OR t LIKE '% fast service %'
        OR t LIKE '% came out quickly %'
        OR t LIKE '% came quickly %'
        THEN 1
    ELSE 0
    END AS wait_pos
/
-- Wait negative
CASE
    WHEN t LIKE '% long wait %'
        OR t LIKE '% waited %'
        OR t LIKE '% waiting %'
        OR t LIKE '% wait time %'
        OR t LIKE '% line out the door %'
        OR t LIKE '% long line %'
        OR t LIKE '% took forever %'
        OR t LIKE '% forever %'
        OR t LIKE '% over an hour %'
        OR t LIKE '% 20 minutes %'
        OR t LIKE '% 30 minutes %'
        OR t LIKE '% 45 minutes %'
        OR t LIKE '% minutes to get %'
        THEN 1
    ELSE 0
    END AS wait_neg
/
-- Cleanliness positive
CASE
    WHEN t LIKE '% clean %'
        OR t LIKE '% very clean %'
        OR t LIKE '% super clean %'
        OR t LIKE '% spotless %'

```

```

OR t LIKE '% immaculate %'
OR t LIKE '% clean bathroom %'
OR t LIKE '% clean restroom %'
OR t LIKE '% clean place %'
OR t LIKE '% clean restaurant %'
THEN 1
ELSE 0
END AS cleanliness_pos
/
-- Cleanliness negative
CASE
WHEN t LIKE '% dirty %'
OR t LIKE '% filthy %'
OR t LIKE '% disgusting %'
OR t LIKE '% gross %'
OR t LIKE '% unclean %'
OR t LIKE '% dirty bathroom %'
OR t LIKE '% dirty restroom %'
OR t LIKE '% bathroom was dirty %'
OR t LIKE '% restroom was dirty %'
OR t LIKE '% hair in %'
OR t LIKE '% hair on %'
OR t LIKE '% bug %'
OR t LIKE '% bugs %'
OR t LIKE '% roach %'
OR t LIKE '% roaches %'
OR t LIKE '% mold %'
OR t LIKE '% smelly %'
OR t LIKE '% bad smell %'
OR t LIKE '% smelled bad %'
OR t LIKE '% odor %'
THEN 1
ELSE 0
END AS cleanliness_neg
/
-- Food positive
CASE
WHEN t LIKE '% delicious %'
OR t LIKE '% tasty %'
OR t LIKE '% flavorful %'
OR t LIKE '% flavourful %'
OR t LIKE '% fresh %'
OR t LIKE '% juicy %'
OR t LIKE '% tender %'
OR t LIKE '% crispy %'
OR t LIKE '% well seasoned %'
OR t LIKE '% perfectly cooked %'
OR t LIKE '% cooked perfectly %'
OR t LIKE '% amazing food %'
OR t LIKE '% great food %'
OR t LIKE '% good food %'
OR t LIKE '% excellent food %'
OR t LIKE '% loved the food %'
THEN 1
ELSE 0
END AS food_pos
/
-- Food negative
CASE
WHEN t LIKE '% bland %'
OR t LIKE '% too salty %'
OR t LIKE '% salty %'
OR t LIKE '% flavorless %'
OR t LIKE '% flavourless %'
OR t LIKE '% tasteless %'
OR t LIKE '% stale %'
OR t LIKE '% greasy %'
OR t LIKE '% food was cold %'
OR t LIKE '% cold food %'
OR t LIKE '% served cold %'
OR t LIKE '% lukewarm %'
OR t LIKE '% overcooked %'

```

```

OR t LIKE '% undercooked %'
OR t LIKE '% burnt %'
OR t LIKE '% raw %'
OR t LIKE '% soggy %'
OR t LIKE '% dry %'
THEN 1
ELSE 0
END AS food_neg
/
-- Value positive
CASE
WHEN t LIKE '% big portion %'
OR t LIKE '% big portions %'
OR t LIKE '% large portion %'
OR t LIKE '% large portions %'
OR t LIKE '% huge portion %'
OR t LIKE '% huge portions %'
OR t LIKE '% generous portion %'
OR t LIKE '% generous portions %'
OR t LIKE '% good portion %'
OR t LIKE '% good portions %'
OR t LIKE '% filling %'
OR t LIKE '% plenty of food %'
THEN 1
ELSE 0
END AS value_pos
/
-- Value negative
CASE
WHEN t LIKE '% small portion %'
OR t LIKE '% small portions %'
OR t LIKE '% tiny portion %'
OR t LIKE '% tiny portions %'
OR t LIKE '% portion was small %'
OR t LIKE '% portions were small %'
OR t LIKE '% not filling %'
OR t LIKE '% skimpy %'
THEN 1
ELSE 0
END AS value_neg
/
-- Environment positive
CASE
WHEN t LIKE '% great atmosphere %'
OR t LIKE '% good atmosphere %'
OR t LIKE '% nice atmosphere %'
OR t LIKE '% cozy atmosphere %'
OR t LIKE '% nice ambience %'
OR t LIKE '% great ambience %'
OR t LIKE '% good vibe %'
OR t LIKE '% nice vibe %'
OR t LIKE '% cozy %'
OR t LIKE '% romantic %'
OR t LIKE '% relaxing %'
OR t LIKE '% comfortable %'
OR t LIKE '% spacious %'
OR t LIKE '% beautiful decor %'
OR t LIKE '% nice decor %'
OR t LIKE '% stylish %'
OR t LIKE '% elegant %'
OR t LIKE '% aesthetic %'
OR t LIKE '% quiet %'
THEN 1
ELSE 0
END AS environment_pos
/
-- Environment negative
CASE
WHEN t LIKE '% too loud %'
OR t LIKE '% loud %'
OR t LIKE '% noisy %'
OR t LIKE '% blasting music %'

```



```

OR t LIKE '% music was loud %'
OR t LIKE '% crowded %'
OR t LIKE '% packed %'
OR t LIKE '% cramped %'
OR t LIKE '% stuffy %'
OR t LIKE '% dark %'
OR t LIKE '% uncomfortable seating %'
OR t LIKE '% bad atmosphere %'
THEN 1
ELSE 0
END AS environment_neg
FROM (
    SELECT r.review_id
           ,r.business_id
           ,r.stars
           ,r.DATE
           ,r.TEXT
           ,CAST(strftime('%Y', r.DATE) AS INTEGER) AS Year
           ,' ' ||
trim(replace(replace(replace(replace(replace(replace(replace(replace(replace(
e(replace(replace(lower(COALESCE(r.TEXT, '')) CHAR(10), ' '), CHAR(13), ' '),
CHAR(9), ' '), ' '), ' '), ' '), ' '), ' '), ' '), ' '), ' '), ' '), ' '), ' '),
'(' , ' ') , ' ') , ' '), '/', ' ')) || ' ' AS t
        FROM review r
        INNER JOIN business b ON r.business_id = b.business_id
        WHERE lower(COALESCE(b.categories, '')) LIKE '%restaurant%'
              OR lower(COALESCE(b.categories, '')) LIKE '%food%'
    ) base
) tags;

```

## [18] Review tags analysis

```

SELECT
    dimension,
    pos_count,
    ROUND(pos_avg, 3) AS pos_avg,
    neg_count,
    ROUND(neg_avg, 3) AS neg_avg,
    ROUND(pos_avg - neg_avg, 3) AS avg_star_gap
FROM (
    SELECT
        'service' AS dimension,
        (SELECT COUNT(*) FROM review_tags WHERE service_pos = 1) AS pos_count,
        (SELECT AVG(stars) FROM review_tags WHERE service_pos = 1) AS pos_avg,
        (SELECT COUNT(*) FROM review_tags WHERE service_neg = 1) AS neg_count,
        (SELECT AVG(stars) FROM review_tags WHERE service_neg = 1) AS neg_avg

    UNION ALL

    SELECT
        'price',
        (SELECT COUNT(*) FROM review_tags WHERE price_pos = 1),
        (SELECT AVG(stars) FROM review_tags WHERE price_pos = 1),
        (SELECT COUNT(*) FROM review_tags WHERE price_neg = 1),
        (SELECT AVG(stars) FROM review_tags WHERE price_neg = 1)

    UNION ALL

    SELECT
        'wait',
        (SELECT COUNT(*) FROM review_tags WHERE wait_pos = 1),
        (SELECT AVG(stars) FROM review_tags WHERE wait_pos = 1),
        (SELECT COUNT(*) FROM review_tags WHERE wait_neg = 1),
        (SELECT AVG(stars) FROM review_tags WHERE wait_neg = 1)

    UNION ALL

    SELECT
        'cleanliness',
        (SELECT COUNT(*) FROM review_tags WHERE cleanliness_pos = 1),
        (SELECT AVG(stars) FROM review_tags WHERE cleanliness_pos = 1),
        (SELECT COUNT(*) FROM review_tags WHERE cleanliness_neg = 1),
        (SELECT AVG(stars) FROM review_tags WHERE cleanliness_neg = 1)

    UNION ALL

    SELECT

```

```

    'food',
    (SELECT COUNT(*) FROM review_tags WHERE food_pos = 1),
    (SELECT AVG(stars) FROM review_tags WHERE food_pos = 1),
    (SELECT COUNT(*) FROM review_tags WHERE food_neg = 1),
    (SELECT AVG(stars) FROM review_tags WHERE food_neg = 1)

UNION ALL
SELECT
    'value',
    (SELECT COUNT(*) FROM review_tags WHERE value_pos = 1),
    (SELECT AVG(stars) FROM review_tags WHERE value_pos = 1),
    (SELECT COUNT(*) FROM review_tags WHERE value_neg = 1),
    (SELECT AVG(stars) FROM review_tags WHERE value_neg = 1)

UNION ALL
SELECT
    'environment',
    (SELECT COUNT(*) FROM review_tags WHERE environment_pos = 1),
    (SELECT AVG(stars) FROM review_tags WHERE environment_pos = 1),
    (SELECT COUNT(*) FROM review_tags WHERE environment_neg = 1),
    (SELECT AVG(stars) FROM review_tags WHERE environment_neg = 1)

) x
ORDER BY avg_star_gap DESC;

```